EGAS-Net: Entropy-Guided Adaptive Selection Network for Hyperspectral–LiDAR Classification

Yuqing Zhao, Yu Song, Chao Liang, Kang Ni, *Member, IEEE*, and Peng Wang, *Senior Member, IEEE*

***Abstract*—The complementarity of hyperspectral imagery (HSI) and light detection and ranging (LiDAR) data provides significant advantages in land-cover classification. In current methods, the cropping size is typically determined by empirical selection, which readily results in the redundancy or loss of land-cover feature statistics. Information entropy, a key measure of data uncertainty and richness, plays a crucial role in selecting feature channels with abundant texture cues and strong discriminability. Motivated by this, this article proposes an entropy-guided adaptive selection network (EGAS-Net) for hyperspectral–LiDAR classification. With information entropy as the core, this method adaptively selects the optimal size for each land cover by calculating spatial information entropy and spatial–spectral information entropy to maximize information content. Then, the spatial and frequency information entropies of HSI and LiDAR data are obtained after cropping. Adaptive weighting is implemented based on the information content ratio of each entropy feature to establish a cross-modal spatial–frequency information interaction mechanism. During the feature fusion stage, the final features are generated through entropy feature weighted coupling to fully exploit the complementary value of HSI–LiDAR data. Experimental results on the Houston 2013, MUUFL, Augsburg, and Yancheng datasets show that the proposed method outperforms other empirical size-based networks, achieving higher classification scores. The code of this project will be available at https://github.com/ZhaoYuQing01/EGASNet**

***Index Terms*—Deep learning, frequency-domain learning, hyperspectral image, information entropy, multisource remote sensing data.**

I. Introduction

HYPERSPECTRAL imagery (HSI) and light detection and ranging (LiDAR) are two typical types of multisource remote sensing data that exhibit remarkable complementary advantages for land-cover classification [1], [2]. With high spectral resolution, HSI can finely characterize the spectral response properties of land covers across different wavelengths, thus facilitating the differentiation of land-cover classes with distinct spectral signatures [3], [4] [5], [6]. In contrast, LiDAR data offers stable and reliable 3-D spatial structure and elevation information, providing unique advantages in describing the geometric morphology of objects, architectural structures, and topographical relief [7], [8]. However, due to inherent differences in physical properties and data dimensions between HSI and LiDAR data, effectively utilizing both data sources for cross-modal feature fusion remains a major challenge for their synergistic classification.

With the advancement of deep learning, there has been increasing interest in fusing spatial- and frequency-domain features for HSI and LiDAR data to fully exploit local spatial details and global frequency-domain structures. Ni et al. [9] proposed FDNet, which constructs a self-attention mechanism based on the fast Fourier transform and integrates a convolutional offset embedding Transformer architecture to adaptively capture local frequency features. Qin et al. [10] proposed dynamic multiple fractional Fourier domains fusion (DMFraF) for the joint classification of hyperspectral and LiDAR data. This method transforms features from both modalities into multiple fractional domains, each containing distinct spatiotemporal frequency components, to achieve multimodal fusion. Kong et al. [11] developed a fast Fourier transform-based frequency-domain optimization self-attention solver to enhance efficiency, enabling effective representation of locally fused features before global frequency-domain learning.

In existing joint classification methods for HSI and LiDAR, fixed-size feature learning remains the predominant paradigm. Xu et al. [12] explored and integrated complementary information from both data sources through two strategies, namely contrastive learning and hierarchical adaptive fusion. Jiao et al. [13] achieved more effective multimodal fusion by distinguishing and extracting common and unique features from the two data sources and establishing dynamic dependencies between them. However, these methods struggle to fully adapt to the significant spatial scale variations of different land covers. To address this issue, some studies have attempted to enhance model adaptability to multiscale objects through multiscale input approaches [14], [15]. Gao et al. [16] improved the joint classification performance of HSI and LiDAR data via spectral feature mining and an adaptive multisource feature fusion mechanism. Shi et al. [17] obtained rich multilevel feature representations through multiscale feature extraction

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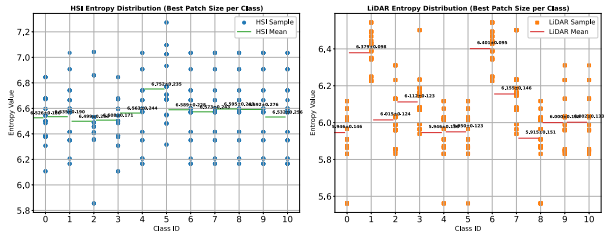


Fig. 1. Information entropy.

and utilized adaptive weight assignment to achieve dynamic cross-modal feature integration. Sun et al. [18] employed a multiscale 3-D–2-D hybrid CNN architecture to extract spatial–spectral features from HSI and elevation features from depth-based LiDAR data. However, the traditional fixed-size method inevitably faces a tradeoff between information loss and redundancy. An undersized scale can truncate crucial spatial topological dependencies of local or fine-grained land covers, resulting in information insufficiency. Conversely, an oversized scale introduces heterogeneous background noise, saturating the multimodal features with statistical redundancy.

To address the limitation that existing studies have paid relatively insufficient attention to the informational validity and uncertainty inherent in the HSI and LiDAR data, we adopt information entropy as the metric to quantify the information content of HSI and LiDAR patches. Information entropy [19], [20], a fundamental concept in information theory for measuring uncertainty, complexity, and information content, can statistically reflect the richness of feature distributions. The advantage of information entropy lies in its nonparametric capability, enabling the quantitative indexing of richness and distributional stability across multimodal attributes. By constructing an entropy maximization constraint, the optimal category-specific context size can be mathematically and precisely localized. This effectively minimizes systemic information loss while simultaneously safeguarding the network against redundant background noise. Fig. 1 shows the information quantity of each category contained in HSI and LiDAR data after adaptively cropping to the optimal size guided by information entropy on the MUUFL dataset. It can be observed that the average information entropy of each class after cropping remains at a high level. This indicates richer information content and more distinct feature differences, which can endow the model with stronger discriminative capability. Furthermore, the samples exhibit favorable clustering characteristics, with more consistent intraclass feature distributions and more stable data distributions, thereby enhancing the robustness of the adaptive patch-size selection strategy. From the perspective of information theory, the proposed method effectively characterizes the discriminative potential of data across different modalities, spatial regions, and scales.

Adaptive adjustment of the feature extraction and selection process according to the complexity of different land-cover types is a key focus of current research. Traditional methods typically rely on fixed-size cropping techniques, which make it difficult to effectively capture the diverse spatial and spectral feature differences between categories. When handling multimodal data, the differences between spatial domain and frequency-domain features are particularly significant. To address this problem, this article proposes an information entropy-based (EB) adaptive cropping method to maximize the discriminative ability of the input data. Regarding the feature fusion problem across different modalities, traditional simple concatenation methods easily introduce redundant information. Therefore, we further propose an information EB adaptive fusion strategy, which guides the adaptive weighted fusion of spatial and frequency-domain features to ensure that features with richer information receive higher weights. By combining adaptive cropping and adaptive fusion techniques, this method effectively enhances the feature representation capability of multimodal data.

This article proposes an entropy-guided adaptive selection network (EGAS-Net) for hyperspectral–LiDAR classification. First, this method employs information entropy as a metric to measure information content. By quantitatively assessing the effective information at different sizes, it adaptively selects the optimal cropping scale for each category, thus enhancing the discriminability of input samples. Based on this, information entropy features from both HSI and LiDAR are learned separately from the spatial and frequency domains, comprehensively capturing the structural complexity and frequency distribution characteristics of the data in different domains. Information entropy is then used to guide feature weighting and coupling, enabling the adaptive fusion of multimodal spatial–frequency statistics. The innovations of this method are as follows.

1) We introduce a plug-and-play entropy-guided size selector (EGSS) that adaptively selects input patch sizes according to the spectral and spatial entropy of different land-cover types, thereby directing the model's attention toward information-rich regions.
2) We propose an EB dual-domain feature module (EB-DDFM). Incorporating the entropy information of spatial and frequency-domain features, this module enhances the fusion performance of multimodal data and effectively improves the model's discriminative capability via adaptive weighting and feature coupling strategies.
3) We propose a learning mechanism that cascades entropy-guided patch size selection with EB spatial–frequency feature fusion. Through module generalization experiments, we validate the transferability of the proposed module to other related algorithms.

The structure of this article is as follows. Section III details the proposed EGAS-Net. Section IV gives the experimental results and analysis. Section V gives the conclusion.

II. Related Work

A. Information Entropy

As a fundamental tool for quantifying the complexity, quality, and uncertainty of data information, information entropy has led to significant research outcomes across various fields. In the field of anomaly detection, Tu et al. [21] combined information entropy with ensemble learning algorithms to process

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hyperspectral data, thereby realizing hyperspectral anomaly detection. The information entropy estimation based on point-set topology (IEEPST) [22] analyzes the statistical properties of hyperspectral data by combining point-set topology with information entropy, striking a balance between computational efficiency and detection accuracy. In the field of semantic segmentation, to address the incorrect mapping of well-aligned features caused by global marginal distribution alignment, Zheng et al. [23] proposed a novel entropy-guided adversarial learning algorithm for aerial image semantic segmentation. In the field of image registration, the minimum relative motion entropy (MRME) [24] achieves matching by constructing the relative motion relationship between candidate matching points and their K nearest neighbors.

Although information entropy has achieved some success in hyperspectral anomaly detection, domain-adaptive semantic segmentation, registration, and scene classification, its systematic application in the task of joint classification of hyperspectral images and LiDAR data is still very rare. Currently, the field of HSI and LiDAR joint classification focuses more on directions, such as deep learning feature fusion, with less use of entropy as a theoretical tool to characterize the information quality, uncertainty, and complementarity between cross-source data.

B. HSI and LiDAR Joint Classification

Spatial frequency feature fusion, which leverages the advantages of collaborative dual-modal spatial details and global dependencies in the frequency domain, has become a key approach for joint classification of HSI and LiDAR. Researchers have explored bimodal complementary information and frequency-domain feature fusion methods using a variety of frequency-domain tools. MCFNet [25] adopts discrete wavelet transform to extract frequency-domain features, captures spatial features through convolutional operations, and enhances modal interaction via an HSI-guided cross-domain fusion module. Both Yang et al. [26] multifrequency graph convolutional network and Wei et al. [27] adaptive frequency-domain sparse enhancement architecture adopt gating mechanisms to explore modal complementarity, solving the problems of insufficient feature mining and noise interference, respectively. In recent years, a variety of cross-modal frequency-domain fusion mechanisms have been widely applied, including networks such as ChirpFAN [28], AF2DN [29], and FHPF-Net [30]. Global–local feature collaborative modeling focuses on high-frequency information retention and detail enhancement. IDNet [31] embeds Laplacian high-frequency features into transformers and mines local–global high-frequency features through a dynamic range perception module. Ma et al. [32] strengthened multiscale feature interaction with a pixel-guided spatial enhancement and frequency-domain collaboration module. Lin et al. [33] balanced global representation and local high-frequency feature capture through a dual-branch structure and a lightweight CNN.

C. Spatial–Frequency Feature Fusion

Relying on the complementary advantages of spatial-domain detail representation and frequency-domain global dependency capture, spatial–frequency feature fusion has been widely explored and applied in multiple subtasks, such as dehazing, super-resolution reconstruction, change detection, image matching, and semantic segmentation [34], [35], [36], [37] [38], [39]. Differentiated technical solutions have been formed for the unique challenges of each scenario. In response to the unique challenges of different application scenarios, the academic community has developed diversified technical solutions. Classified by methodological categories, existing studies can be mainly summarized into two core directions: frequency-domain information enhancement and feature representation methods, and spatial–frequency interaction and attention optimization mechanisms.

The core objective of frequency-domain information enhancement and feature representation optimization is to deeply mine and optimize frequency-domain feature information, thereby improving the quality of feature expression. The Transformer-guided CycleGAN framework [40] utilizes Fourier-domain correlation analysis to enhance target recognition after image dehazing. Hou et al. [41] further improved frequency representation through amplitude–phase learning and frequency-domain loss optimization. In addition, methods such as the MDCTNet [42] and OFLP [43] have laid a foundation for subsequent spatial–frequency information enhancement of cross-domain images. Spatial–frequency interaction mechanisms focus on adaptive information exchange between spatial and frequency domains. The boundary-aware spatial and frequency dual-domain Transformer [44] achieves spatial–temporal feature complementarity via a dual-domain mixer and enhances boundary detail capture capability by combining a boundary-guided training strategy. Hui et al. [45] proposed an adaptive frequency–spatial interaction attention mechanism, which dynamically generates frequency-domain weights through a head-specific lightweight frequency-domain projection network; this mechanism conducts adaptive interaction with the spatial attention mechanism to implement frequency-domain-guided spatial feature learning.

Although existing studies have demonstrated the effectiveness of spatial–frequency feature fusion, adaptively acquiring spatial and frequency-domain representations while considering feature redundancy in HSI and LiDAR data contributes to a substantial improvement in the accuracy of collaborative HSI and LiDAR classification.

III. Methodology

As illustrated in Fig. 2, the EGAS-Net is specifically designed for the joint classification of hyperspectral image (with multiple spectral channels) and LiDAR point cloud data [represented as digital surface models (DSMs), of absolute heights above sea level]. Information entropy offers an interpretable metric for quantifying intrinsic information richness, and it can facilitate adaptive selection across diverse levels, including spatial context scales, feature contributions, and frequency representations.

A. EGSS

In the HSI and LiDAR joint classification task, the cropping size directly affects the model's capability to perceive local

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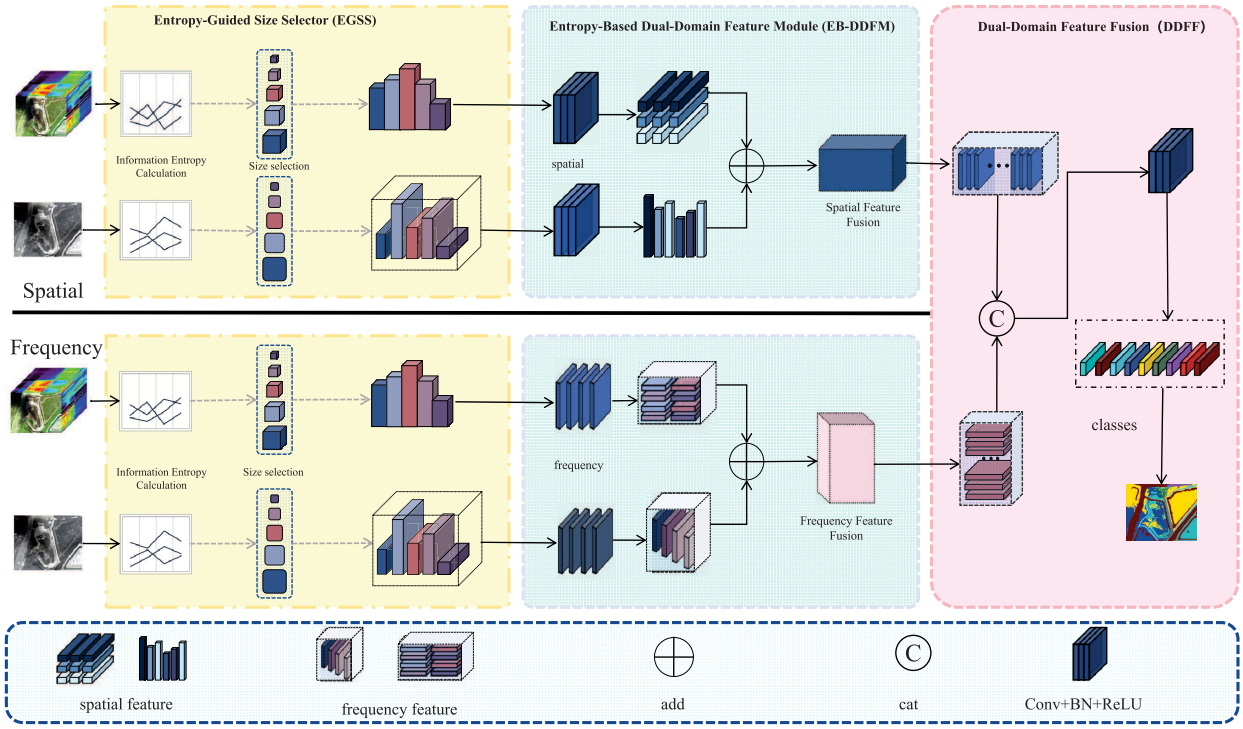


Fig. 2. Architecture of EGAS-Net.

details and contextual information. An excessively small size may result in the loss of critical spatial context, while an excessively large size can introduce redundant background noise. Most existing methods rely on conventional experience to select a fixed cropping size, which fails to adapt to the complex structural characteristics of different land-cover types or regions. To address this issue, this article introduces information entropy as a quantitative evaluation metric before model input, as illustrated in Fig. 3. Given the distinct characteristics of HSI and LiDAR data, spatial–spectral information entropy and spatial information entropy are extracted separately. By quantifying the information richness of input data at different cropping sizes, the optimal cropping size is adaptively selected for each land-cover class, which thus enhances the discriminability of input samples.

First, the input data is preprocessed. The HSI undergoes radiometric and atmospheric corrections to remove environmental noise, with a spectral dimension of L and spatial resolution of $H \times W$. The LiDAR data is rasterized into an $H \times W$ 2-D digital elevation image after denoising. Information entropy is defined based on Shannon entropy, with the core formula given as

$$H = -\sum_{x \in \boldsymbol{X}} p(x) \log_2 p(x) \tag{1}$$

where $\boldsymbol{X}$ denotes the set of data values, and $p(x)$ represents the probability distribution of x, calculated by normalizing the data histogram. $h(x)$ is the frequency of x in the histogram, that is, $p(x) = h(x)/\Sigma_{x \in X} h(x)$ (where $h(x)$ is the frequency of x in the histogram). A higher entropy value H indicates that the data contains richer effective information. Considering the "spatial-spectral" 3-D characteristics of HSI, the gray-level distribution of the data volume is calculated and substituted into the Shannon entropy formula to obtain the joint information entropy of HSI

$$H_{HSI}(k) = -\sum_{i=1}^{k}\sum_{j=1}^{k}\sum_{l=1}^{L} p(\boldsymbol{I}_{3D}(i,j,l)) \log_2 p(\boldsymbol{I}_{3D}(i,j,l)). \tag{2}$$

For LiDAR data, the quantitative distribution in the spatial dimension is statistically analyzed to calculate its information entropy

$$H_{LiDAR}(k) = -\sum_{i=1}^{k}\sum_{j=1}^{k} p(\boldsymbol{I}_{2D}(i,j)) \log_2 p(\boldsymbol{I}_{2D}(i,j)). \tag{3}$$

The optimal patch size should balance information richness and statistical stability. Although larger entropy values generally indicate more abundant discriminative information, maximizing entropy alone may introduce excessive background regions and redundant information. Therefore, the entropy standard deviation is incorporated as a stability constraint. Combining mean entropy and entropy standard deviation, the selected patch size retains valuable spatial–spectral features and ensures stable statistical properties

$$\bar{H}(c) = \frac{1}{N}\sum_{n=1}^{N} H(c,n) \tag{4}$$

$$\sigma_H(c) = \sqrt{\frac{1}{N}\sum\nolimits_{n=1}^{N} \left(H(c,n) - \bar{H}(c)\right)^2} \tag{5}$$

where $\bar{H}(c)$ calculates the average information entropy of all samples of class c, used to determine the most stable

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information content of the class. $\sigma_H(c)$ calculates the standard deviation; the smaller the standard deviation, the more stable the class information, and the less noise at that size. The aforementioned computation is implemented in data preprocessing without introducing any trainable parameters. Entropy computation is completed before network training, bringing nearly no extra computational cost in the inference stage

$$k^* = \arg\max_k \left\{ \bar{H}(c,k) \mid \sigma_H(c,k) < \sigma_H(c,k_{\text{best}}) \right\} \tag{6}$$

where $\arg\max_k$ selects the candidate patch size k maximizing the objective, $\bar{H}(c,k)$ is the mean entropy of class c at size k, $\sigma_H(c,k)$ is the entropy standard deviation, and k_{best} is the current optimal size.

Based on information entropy theory, the optimal patch size should balance information richness and statistical stability. Although larger entropy values generally indicate more abundant discriminative information, maximizing entropy alone may introduce excessive background regions and redundant information. Therefore, the entropy standard deviation is incorporated as a stability constraint to evaluate the consistency of information distribution within the same class. By jointly maximizing the mean entropy while minimizing its variation, the selected patch size preserves the most informative spatial–spectral characteristics while maintaining stable statistical properties, thereby approximating the optimal context size for each land-cover category. It should be noted that LiDAR-derived DSM data may inevitably contain elevation noise or missing values due to sensor errors, occlusion, and acquisition limitations. Since EGSS determines the optimal context size based on entropy statistics rather than individual pixels, random elevation perturbations only introduce local variations in entropy estimation. Moreover, the proposed entropy standard deviation constraint evaluates the stability of information distribution, preventing abnormally high-entropy regions caused by noise from being incorrectly selected as optimal scales.

This approach differs from the previous method of directly selecting a uniform cropping size; instead, the optimal size for each category is determined based on its unique characteristics. By synthesizing the information content of samples from different categories, the optimal cropping size for each category is selected for category-specific cropping. This ensures that the local samples input to the model contain as rich and effective discriminative information as possible at the current scale, thereby improving the reliability of subsequent feature extraction and fusion.

B. EB-DDFM

Hyperspectral images and LiDAR are heterogeneous multimodal data, with significant differences in representation domains and information dimensions. Direct fusion can easily introduce redundancy and reduce discriminability. To address this, this section constructs an EB-DDFM, which combines dual-domain information entropy feature extraction with entropy-guided adaptive feature coupling, as shown in Fig. 4. It explores structural and information distribution features in both spatial and frequency domains and quantifies modal differences using information entropy. Additionally, adaptive weight allocation is designed using entropy values as a reliability criterion, achieving precise coupling of multimodal dual-domain features, suppressing redundancy, and retaining effective information, laying the foundation for subsequent high-precision learning and inference.

To comprehensively characterize the structural complexity of HSI and LiDAR data across different representation domains, this article first extracts information entropy features from two complementary perspectives: the spatial domain and the frequency domain. Spatial domain information entropy reflects the distribution differences and structural complexity of pixels within local neighborhoods, while frequency-domain information entropy reveals the energy distribution characteristics of textural details and structural changes in the frequency dimension. For HSI, the extraction of spatial 3-D entropy H_{HS} follows the logic of 'spatial–spectral' joint entropy. It is obtained by counting the gray-level distribution of the 3-D data volume composed of HSI in both spatial and spectral dimensions and substituting it into the Shannon entropy formula. This feature accurately characterizes the spatial correlation and spectral variability between HSI pixels, preserving the structural information of the original data. The extraction of frequency-domain 3-D entropy H_{HF} requires prior frequency-domain transformation: first, 2-D fast Fourier transform is applied to each spectral dimension of HSI to convert the spatial-domain data into a frequency-domain complex matrix, where $FFT_2()$ denotes the 2-D fast Fourier transform operator. Subsequently, the magnitude of the complex matrix is extracted to eliminate the phase information, which is susceptible to noise interference, obtaining the frequency-domain magnitude matrix $|F_l|$. Finally, the gray-level distribution of the frequency-domain 3-D data volume, composed of the frequency-domain magnitude matrices of all spectral dimensions, is counted, and the frequency-domain 3-D entropy is calculated as follows:

$$\boldsymbol{F}_l(u,v) = FFT2(\boldsymbol{I}_l(i,j)) \tag{7}$$

$$\left|\boldsymbol{F}_l(i,j)\right| = \sqrt{\mathrm{Re}(\boldsymbol{F}_l(i,j))^2 + \mathrm{Im}(\boldsymbol{F}_l(i,j))^2} \tag{8}$$

where $\boldsymbol{F}_l(u,v)$ denotes the complex-valued frequency matrix. The frequency magnitude is then extracted to eliminate noise-sensitive phase information. The probability distribution of the magnitude values is calculated as

$$P(|\boldsymbol{F}_l(i,j)|) = \frac{h(|\boldsymbol{F}_l(i,j)|)}{\sum_{i,j,l} h(|\boldsymbol{F}_l(i,j)|)} \tag{9}$$

$$H_{HF} = -\sum_{i=1}^{k}\sum_{j=1}^{k}\sum_{l=1}^{L} P(|\boldsymbol{F}_l(i,j)|)\, log_2 P(|\boldsymbol{F}_l(i,j)|). \tag{10}$$

This feature effectively captures both high-frequency details and low-frequency global information of HSI. For LiDAR, the spatial entropy H_{LS} is calculated by counting the quantitative distribution of elevation values, which characterizes the spatial heterogeneity of LiDAR elevation information (e.g., terrain undulations and elevation differences between building edges and the ground). The extraction process for frequency-domain

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entropy H_{LF} follows the same procedure as that for HSI frequency-domain entropy: first, perform 2-D-FFT on LiDAR to obtain a frequency-domain complex matrix; after extracting the magnitude, count its gray-level distribution and calculate the frequency-domain entropy as

$$H_{LF} = -\sum_{i=1}^{k}\sum_{j=1}^{k} P\left(|\boldsymbol{F}_L(i,j)|\right) log_2 P\left(|\boldsymbol{F}_L(i,j)|\right). \tag{11}$$

This feature captures the frequency-domain energy distribution of LiDAR elevation information, such as the concentration of low-frequency energy in large-scale flat areas and the peaks of high-frequency energy in regions with abrupt terrain changes.

In the process of multimodal feature fusion, directly concatenating features from different modalities can introduce redundant information and even degrade discriminative capability. To address this issue, this article introduces information entropy as a metric for measuring feature reliability and importance, and constructs an information entropy-guided adaptive feature coupling mechanism, where features with richer information content are assigned higher weights during the fusion process.

To fully exploit the complementarity between each entropy feature and local spatial features, and avoid the blindness of feature fusion, this module realizes the enhancement of effective information and the suppression of redundant information through adaptive weight assignment and feature coupling strategies. Higher entropy indicates richer structural variability and stronger information diversity. Therefore, entropy is used as a reliability indicator to allocate larger contributions to more informative modalities. First, the HSI and LiDAR information entropies obtained in the spatial and frequency domains are subjected to normalization, and adaptive weight coefficients are generated accordingly. Subsequently, the weighted HSI spatial entropy is multiplied with its corresponding spatial features in an element-wise manner, while the weighted LiDAR spatial entropy is fused with LiDAR spatial features, thereby achieving entropy-guided enhancement of spatial structural information. In the frequency domain, the same strategy is adopted: the frequency-domain entropies of HSI and LiDAR are, respectively, coupled with their corresponding frequency-domain features in a weighted manner, enabling the frequency energy distribution to participate in feature representation more reasonably

$$w_{AS} = \frac{H_{AS}}{H_{HS} + H_{LS}}, w_{AF} = \frac{H_{AF}}{H_{HF} + H_{LF}} \tag{12}$$

where A represents H or L. H_{HF} denotes the frequency-domain entropy of the HSI branch and H_{LF} denotes the frequency-domain entropy of the LiDAR branch. H_{HS} represents the spatial information entropy of the HSI branch, and H_{LS} represents the spatial information entropy of the LiDAR branch

$$\boldsymbol{F}_S = w_{HS} \cdot \boldsymbol{F}_{HS} + w_{LS} \cdot \boldsymbol{F}_{LS} \tag{13}$$

$$\boldsymbol{F}_F = w_{HF} \cdot \boldsymbol{F}_{HF} + w_{LF} \cdot \boldsymbol{F}_{LF} \tag{14}$$

where $\boldsymbol{F}_{HF}$ represents the frequency features of the HSI branch, and $\boldsymbol{F}_{HS}$ represents the spatial features of the HSI branch. $\boldsymbol{F}_{LF}$ represents the frequency features of the LiDAR branch, and $\boldsymbol{F}_{LS}$ represents the spatial features of the LiDAR branch.

This coupling mechanism avoids the uncertainty caused by the manual setting of fusion weights. By leveraging information entropy to adaptively regulate the contribution ratios of features from different modalities and different representation domains, it effectively suppresses redundant information and highlights highly discriminative structural and texture features. Through the weight guidance of spatial and frequency features by entropy features, this strategy strengthens effective information, suppresses redundant information, and improves the discriminative capability of the features.

C. Dual-Domain Feature Fusion

After completing the information entropy-guided spatial and frequency-domain feature coupling, the fused multimodal spatial–frequency features are concatenated into a unified high-dimensional feature representation, which is then input into the classification network for final discrimination. This feature incorporates the spectral–spatial structural information of HSI, the elevation geometric characteristics of LiDAR, and the complementary texture and structural representations of both in the frequency domain

$$F_{global} = \mathrm{ReLU}\left(\mathcal{N}\left(\mathcal{C}\left(\mathrm{Concat}\left(\boldsymbol{F}_S, \boldsymbol{F}_F\right)\right)\right)\right) \tag{15}$$

$$\boldsymbol{Out} = F_{global} \cdot \boldsymbol{w}_{cls}^T + b_{cls} \tag{16}$$

where $\mathcal{C}(\cdot)$ denotes the convolution operation of the fusion convolutional layer, $\mathcal{N}(\cdot)$ denotes the normalization layer, and ReLU is the rectified linear unit activation function.

By introducing information entropy constraints at the feature level, the resulting fused features not only preserve multimodal complementarity but also effectively mitigate the interference of background noise and redundant information, thereby enhancing interclass separability. This module significantly improves the stability and overall accuracy (OA) of joint classification, validating the effectiveness of the information entropy-driven spatial–frequency fusion strategy.

IV. Experiments

A. Dataset

In this experiment, four commonly used HSI and LiDAR datasets were used to evaluate the effectiveness of our proposed model.

1) Houston2013 Dataset: This dataset [15], [46] encompasses hyperspectral scenes of the University of Houston campus and its surrounding urban regions, integrating HSI- and LiDAR-derived DSM data. Both modalities feature a uniform spatial dimension of 349 $\times$ 1905 pixels and a ground sampling distance of 2.5 m. The HSI component includes 144 spectral channels spanning wavelengths from 0.38 to 1.05 mm, while the LiDAR data consists of a single-band elevation representation. Altogether, the dataset comprises 15 029 labeled ground-truth samples, systematically categorizing 15 different land-cover types for analytical purposes.

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TABLE I
COMPARISON OF THE CLASSIFICATION ACCURACIES USING THE HOUSTON 2013 DATASET

No.	Class (Train/Test)	MHST	HCT-Net	DFFNet	FDNet	MCFNet	MSA-GCN	M2FNet	mPMCL	MMD-MLP	EGAS
1	Health Grass (20/1231)	98.05	89.03	90.08	97.32	93.83	90.09	94.15	**99.35**	84.97	98.13
2	Stressed Grass (20/1234)	98.22	96.52	78.03	98.78	96.55	90.36	82.33	90.28	80.31	**99.03**
3	Synthetic Grass (20/677)	99.26	**99.85**	99.26	97.49	99.69	98.08	**99.85**	98.52	97.78	98.97
4	Trees (20/1224)	99.59	99.51	80.31	**99.84**	97.64	99.26	91.75	94.36	99.59	99.59
5	Soil (20/1222)	99.35	99.92	99.18	**100**	99.99	99.59	99.84	**100**	99.18	99.59
6	Water (20/305)	99.67	**100**	92.13	96.39	96.85	91.80	**100**	98.69	93.11	99.76
7	Residential (20/1248)	95.59	99.04	70.27	96.23	95.09	64.66	96.55	97.36	61.38	**100**
8	Commercial (20/1224)	90.77	89.38	89.21	89.54	89.22	79.08	91.99	78.35	61.60	**99.67**
9	Road (20/1232)	89.20	84.01	62.25	84.50	**93.99**	47.08	76.54	86.77	59.33	90.02
10	Highway (20/1207)	96.77	92.87	84.17	**98.18**	93.12	49.05	95.03	93.37	57.17	**97.18**
11	Railway (20/1215)	94.32	94.16	96.95	96.30	95.14	50.37	**98.60**	96.54	75.47	94.65
12	Parking Lot 1 (20/1213)	93.82	87.63	84.58	**95.47**	92.08	57.71	81.95	80.87	91.43	93.82
13	Parking Lot 2 (20/449)	97.33	**100**	95.32	91.98	98.15	66.15	95.99	99.33	65.70	97.55
14	Tennis Court (20/408)	**100**	**100**	**100**	**100**	**100**	99.02	**100**	**100**	97.79	**100**
15	Running Track (20/640)	**100**	**100**	**100**	**100**	**100**	**100**	99.84	99.53	**100**	**100**
	OA	96.19	94.02	85.89	95.91	95.43	76.04	92.26	92.99	79.53	**97.52**
	AA	96.80	95.19	88.11	96.13	96.09	78.82	93.62	94.22	81.65	**97.86**
	Kappa	95.88	93.53	84.75	95.58	95.06	74.07	91.64	92.42	77.85	**97.32**

TABLE II
COMPARISON OF THE CLASSIFICATION ACCURACIES USING THE MUUFL DATASET

No.	Class (Train/Test)	MHST	HCT-Net	DFFNet	FDNet	MCFNet	MSA-GCN	M2FNet	mPMCL	MMD-MLP	EGAS
1	Trees (60/23186)	91.81	87.32	76.75	90.60	89.23	87.00	87.34	88.56	86.99	**92.29**
2	Mostly grass (60/4210)	85.70	75.99	67.19	83.28	78.95	73.02	64.58	85.94	84.58	**86.22**
3	Mixed ground surface (60/6822)	72.78	69.29	60.94	75.48	75.49	**77.51**	62.47	77.22	72.10	73.62
4	Dirt and sand (60/1766)	89.81	95.13	94.56	89.01	94.33	88.22	89.52	**95.75**	85.96	90.14
5	Road (60/6627)	88.29	81.67	68.23	87.94	87.32	75.40	82.86	85.23	57.19	**88.74**
6	Water (60/406)	**100**	**100**	**100**	**100**	**100**	**100**	99.75	**100**	97.29	**100**
7	Buildings shadow (60/2173)	93.05	90.29	90.42	92.50	**95.58**	90.70	84.91	89.60	73.45	93.32
8	Buildings (60/6180)	95.52	94.66	91.29	95.87	95.17	94.50	92.61	95.70	93.90	**96.02**
9	Sidewalk (60/1325)	82.19	64.60	38.41	75.62	81.73	42.94	48.98	70.34	43.77	**82.79**
10	Yellow curb (60/123)	92.68	82.93	73.98	93.50	91.86	78.86	75.61	88.62	**94.31**	93.50
11	Cloth panels (60/209)	99.04	**99.52**	99.04	98.56	97.12	98.09	**99.52**	**99.52**	**99.52**	99.04
	OA	88.71	84.20	75.04	88.12	87.46	83.30	81.51	87.26	80.44	**89.22**
	AA	90.08	85.58	78.26	89.31	89.71	82.38	80.74	88.70	80.82	**90.52**
	Kappa	85.32	79.67	68.55	84.58	83.80	78.43	76.13	83.55	74.80	**85.95**

2) MUUFL Dataset: This dataset [47], [48] is acquired by capturing HSI- and LiDAR-derived DSM data over the Gulfport Campus of the University of Southern Mississippi, located in Long Beach, Mississippi, USA. The HSI data is collected using the ITRES CASI-1500 sensor, while the LiDAR data is obtained through an ALTM sensor. The imagery spans dimensions of 325 pixels in width and 220 pixels in height. The HSI comprises 64 spectral bands covering wavelengths from 0.38 to 1.05 μm, with a spatial resolution of 0.54 $\times$ 1.0 m. The LiDAR data, captured by the ALTM sensor utilizing a 1064-nm laser wavelength, offers a spatial resolution of 0.60 by 0.78 m. In total, the dataset encompasses 53 687 ground-truth samples and includes 11 different land covers for analysis.

3) Augsburg Dataset: This dataset [49] comprises synthetic spaceborne hyperspectral imagery of Augsburg, Germany, alongside DSM data—a monochromatic image that provides precise elevation measurements for the observed area. This hyperspectral component of the dataset includes 180 spectral bands, spanning wavelengths from 400 to 2500 nm. The spatial dimensions of the imagery are 332 in width by 485 pixels in height, with a ground sampling distance (spatial resolution) of 30 m.

4) Yancheng Dataset: This dataset [9] was acquired over the urban area of Yancheng, China. It contains both HSI- and LiDAR-based DSM. The data size is 400 $\times$ 500 pixels with a spatial resolution of 1 m. The HSI data were obtained by the visible-light imaging spectrometer of the multiparameter airborne remote sensing system (MARSS), which includes 249 spectral bands with a wavelength range of 400–1000 nm. To reduce redundant information and computational cost, principal component analysis (PCA) was adopted to reduce the number of spectral channels of the HSI data to 30. The LiDAR data were collected by a RIEGL VUX-240 laser scanner with a single band and an average point density of 51.44 points per square meter. The dataset covers 9 distinct land-cover classes with a total of 200 000 real samples.

B. Experimental Setup

The proposed network is implemented based on the PyTorch framework. The experiment was conducted on an Ubuntu 22.04 system, using the AdamW optimizer for network training, and the learning rate decay parameter was set to 0.9. The batch size was set to 32 and the training rounds to 1000 during the training phase. Standard cross-entropy is used as the loss function. To objectively evaluate the classification performance, three quantitative indicators are introduced: OA, average accuracy (AA), and the Kappa coefficient. Tables I–IV show the numbers of training and testing samples used in this study.

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TABLE III
COMPARISON OF THE CLASSIFICATION ACCURACIES USING THE AUGSBURG DATASET

No.	Class (Train/Test)	MHST	HCT-Net	DFFNet	FDNet	MCFNet	MSA-GCN	M2FNet	mPMCL	MMD-MLP	EGAS
1	Forest (20/13487)	99.20	95.37	97.75	98.12	98.06	98.67	96.25	98.29	98.47	**99.32**
2	Residential Area (20/30309)	82.88	65.87	36.81	78.26	78.37	35.99	83.15	87.25	77.21	**89.22**
3	Industrial Area (20/3831)	66.35	39.08	54.16	52.60	61.75	40.12	**77.03**	71.10	61.89	66.43
4	Low Plants (20/26837)	89.70	87.03	74.29	85.39	85.65	65.80	89.07	88.90	72.30	**90.43**
5	Allotment (20/555)	92.25	94.95	**96.03**	82.34	91.89	82.52	89.73	95.14	77.30	92.79
6	Commercial Area (20/1625)	50.64	63.38	31.81	58.71	58.83	31.63	50.03	50.95	53.23	**93.11**
7	Water (20/1510)	58.54	62.05	62.11	**67.75**	62.05	43.58	57.09	65.10	45.23	66.95
	OA	86.16	76.99	61.85	82.30	82.83	57.63	85.99	87.80	77.32	**89.94**
	AA	77.08	72.53	64.71	74.74	76.66	56.90	77.47	79.53	69.37	**85.46**
	Kappa	80.93	69.58	52.19	76.00	76.55	45.53	80.73	83.13	69.40	**85.93**

TABLE IV
COMPARISON OF THE CLASSIFICATION ACCURACIES USING THE YANCHENG DATASET

No.	Class (Train/Test)	MHST	HCT-Net	DFFNet	FDNet	MCFNet	MSA-GCN	M2FNet	mPMCL	MMD-MLP	EGAS
1	Suaeda glauca (20/2520)	91.03	94.56	95.16	**98.25**	86.83	94.40	91.34	94.13	87.74	97.34
2	Bare soil (20/30275)	72.62	24.66	68.36	71.65	70.08	66.55	67.62	**74.42**	63.09	73.22
3	Road (20/2956)	86.71	77.33	87.18	81.22	**89.51**	82.17	85.92	84.88	89.14	82.10
4	Buildings (20/87)	98.85	**100**	98.85	**100**	**100**	82.76	**100**	96.55	**100**	**100**
5	Water (20/49297)	83.31	77.85	76.85	88.18	86.55	**89.45**	72.09	87.24	78.44	88.18
6	Reed (20/42251)	66.29	22.43	66.00	68.97	67.07	69.84	**84.13**	72.15	70.50	69.01
7	S. alterniflora (20/58880)	69.58	**82.81**	66.58	73.80	67.47	51.30	66.08	61.32	57.66	**82.81**
8	Algae (20/3282)	87.11	89.52	86.93	**94.94**	87.11	74.74	88.54	82.39	79.89	87.23
9	Grass (20/10272)	65.21	69.83	67.53	**71.38**	68.70	69.43	64.90	70.31	68.14	70.10
	OA	73.33	59.53	70.32	76.65	73.46	69.27	72.97	73.57	68.09	**79.36**
	AA	80.07	71.00	79.27	83.15	80.36	75.62	80.07	80.37	77.17	**83.33**
	Kappa	66.64	49.20	63.04	71.03	67.03	61.58	66.21	67.31	60.59	**73.60**

C. Performance Comparison

Tables I–IV show the objective classification performance of different methods. In this study, representative models from recent years were selected for comparative analysis, including MHST [50], HCT-Net [49], DFFNet [51], FDNet [9], MCFNet [25], MSA-GCN [52], M2FNet [18], mPMCL [53], and MMD-MLP [14]. These architectures can be broadly divided into two categories: models based on spatial feature learning only and models based on frequency spatial feature learning. The results show that models using frequency spatial feature learning architecture generally have higher accuracy than models that rely solely on spatial feature learning architecture.

Multiple comparison methods were evaluated using visual comparison and quantitative indicators, including OA, AA, and the Kappa coefficient. Tables I–IV present the objective classification results of the proposed model and comparison methods on the Houston2013, MUUFL, Augsburg, and Yancheng datasets, with the best results for each row highlighted in bold.

1) Quantitative Index Analysis: Tables I–IV present the quantitative classification results of various methods on four typical remote sensing datasets: Houston 2013, MUUFL, Augsburg, and Yancheng. Overall, compared to other recent models, the proposed model outperforms all others across the datasets.

For the Houston2013 dataset (Table I), which contains 15 land-cover categories, the following conclusions can be drawn. The EGAS-Net proposed in this article demonstrates significant performance advantages in multimodal HSI–LiDAR classification tasks, ranking first in overall classification accuracy among all compared methods. It not only outperforms mainstream methods, such as MHST, FDNet, and DFFNet, but also achieves high precision and stable outputs across most categories, exhibiting stronger robustness. The model excels particularly in traditionally difficult-to-classify categories. For instance, in the "stressed grass" category, which has highly similar spectral features and easily confusable context, EGAS-Net significantly outperforms methods like DFFNet and MMD-MLP, demonstrating the model's strong ability to distinguish subtle spectral and spatial structural differences. This is primarily credited to EGSS, which adaptively selects the optimal input size based on class-specific information entropy, thereby effectively enhancing the perception of subtle spectral variations. Furthermore, EB-DDFM strengthens the model's discriminative power by assigning higher weights to highly informative spatial–frequency features, which successfully capture the fine-grained structural differences between "stressed grass" and "healthy grass." For artificial land-cover categories with pronounced linear structures, such as "road," "highway," and "railway," EGAS-Net also achieves high accuracy, effectively addressing the severe accuracy collapse observed with methods like MSA-GCN and MMD-MLP in such scenarios. Meanwhile, in small-sample categories, such as "water" and "parking lot 2," EGAS-Net still maintains excellent performance. Benefiting from the information entropy-driven spatial–frequency feature fusion mechanism, this model can fully couple the spectral–spatial information of HSI with the elevation and geometric features of LiDAR, effectively exploring the complementarity between multimodal data. In urban building categories, such as commercial and residential, where background noise is complex and spectral overlap is severe, it achieves accuracy that

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significantly surpasses classification results relying on single modalities. Additionally, through entropy quantification and adaptive weight regulation, it effectively suppresses redundant noise interference and strengthens the expression of highly discriminative structural features. Across all 15 land-cover categories, EGAS maintains a consistently high level of precision, comprehensively verifying the model's excellent overall performance, feature extraction capability, and robustness in multimodal remote sensing data classification tasks.

As shown in Table II, our model outperforms the second-best model by 0.51% in accuracy on the MUUFL dataset. For land-cover types with strong spatial structures, such as "trees" and "buildings," the model consistently obtains superior performance, which can be attributed to the EB-DDFM module. This enables enhanced characterization of heterogeneous vegetation textures and regular geometric structures of artificial objects, thereby improving cross-modal feature consistency. For easily confused categories, such as "sidewalk," which exhibits substantial spatial overlap with roads and building shadows, the EGSS module adaptively adjusts the context size based on information entropy, enhancing local boundary discrimination and resulting in improved classification accuracy. For challenging categories, such as "mostly grass," which exhibit high spectral similarity with other vegetation and substantial intraclass variability, the model maintains stable performance by leveraging spatial–frequency fusion to suppress background interference while capturing statistical fluctuations. Overall, the results demonstrate balanced performance across all classes with no obvious weaknesses, indicating strong generalization ability and robustness of the proposed method in complex multisource remote sensing scenarios.

In the Augsburg dataset test (Table III), our model demonstrated a significant advantage. In the "forest" category, EGSS can adaptively select the optimal context size based on the canopy distribution and spatial structural characteristics of forest areas. It preserves critical contextual information while eliminating interference from redundant backgrounds, thus achieving better classification performance than methods such as MHST. In the "residential area" category, due to the size selection strategy guided by information entropy, the model significantly outperforms models such as DFFNet and MSA-GCN. At the same time, the "low plants" category is highly spectrally similar to green vegetation categories, such as "allotment" and "forest" edges, making them difficult to distinguish solely relying on spectral features. In EB-DDFM, however, the adaptive frequency-domain weighting mechanism can highlight subtle differences in texture density and spatial distribution structure among different vegetation types, thereby achieving more stable and superior performance. In contrast, other models showed clear shortcomings. For example, DFFNet and MSA-GCN had very low accuracy in categories like residential area and commercial area, while models like M2FNet and mPMCL performed reasonably in some categories but lacked overall balance compared to EGAS. In summary, the EGAS-Net consistently maintains stable and excellent performance across all categories, with better generalization and robustness.

From Table IV, it can be seen that on the Yancheng dataset, EGAS-Net's accuracy improved by 2.71% compared to the second place. Benefiting from the entropy-guided spatial–frequency adaptive fusion technique, our method effectively alleviates the confusion between fine-grained features (e.g., "water," "reed," and "S. alterniflora") and complex backgrounds. In contrast, models such as MHST and HCT-Net perform well on several subcategories but lack stability in the highly challenging "S. alterniflora" class. Models such as DFFNet, MCFNet, and MSA-GCN perform significantly worse in most categories. Overall, the EGAS-Net demonstrates stronger feature representation capability and robustness in HSI and LiDAR joint classification tasks.

Fig. 5 plots OA, AA, and Kappa curves of EGAS-Net on Houston2013, MUUFL, Augsburg, and Yancheng under three training sample setups: 1%, 5%, and stepped unequal per-class samples (5/10/15/20). Even with only 1% extremely limited training samples, our model maintains a small gap between OA and AA without bias toward majority land-cover types. Powered by entropy-guided adaptive cropping and balanced multimodal fusion, the network adapts to naturally imbalanced remote sensing data without manual resampling.

Although the four datasets exhibit distinct differences in class distributions, the proposed model achieves stable OA, AA, and Kappa coefficient and obtains favorable classification performance on minority classes. Taking the MUUFL dataset as an example, although there is a huge difference in sample numbers between most classes like trees and minority classes like yellow curb, EGAS-Net achieves classification accuracies of 92.29% and 93.50% for these two minority classes. This robustness mainly originates from the EGSS and EB-DDFM modules. The patch size selection and feature fusion weights are determined by information entropy rather than sample quantity. Consequently, the model is robust to uneven sample distributions among different categories. It avoids overemphasizing features of majority classes simply due to their larger sample proportions. The per-class accuracy of categories with scarce small samples does not degrade obviously, leading to highly close values between AA and OA. Comprehensive quantitative analysis results demonstrate that our model achieves comprehensive leadership in both composite metrics (OA, AA, Kappa) and individual class accuracies through more efficient frequency feature extraction and fusion strategies, fully validating its superiority in complex land-cover classification tasks.

2) Visual Comparative Analysis: Figs. 6–9 show the visual classification results of multiple comparison methods on the Houston2013, MUUFL, Augsburg, and Yancheng datasets. The MUUFL and Augsburg datasets clearly illustrate the performance differences between different models through local magnification. It can be seen that the visual classification results of the proposed method are closer to the actual map, with smoother generated classification results, demonstrating better classification performance. Due to the precise matching of the optimal context size for different land-cover categories by the EGSS during the data preprocessing, the proposed model maximizes the preservation of spatial topological context at a macroscale, overcoming the

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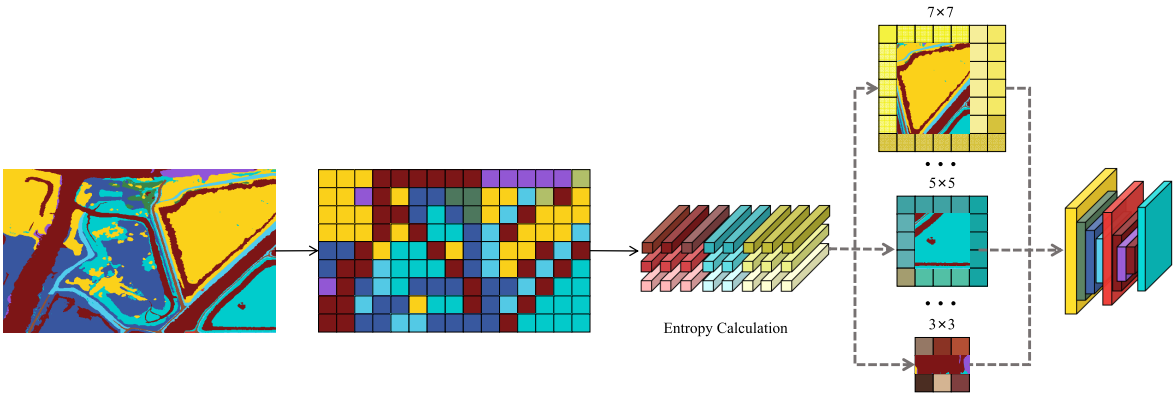


Fig. 3. Architecture of EGSS.

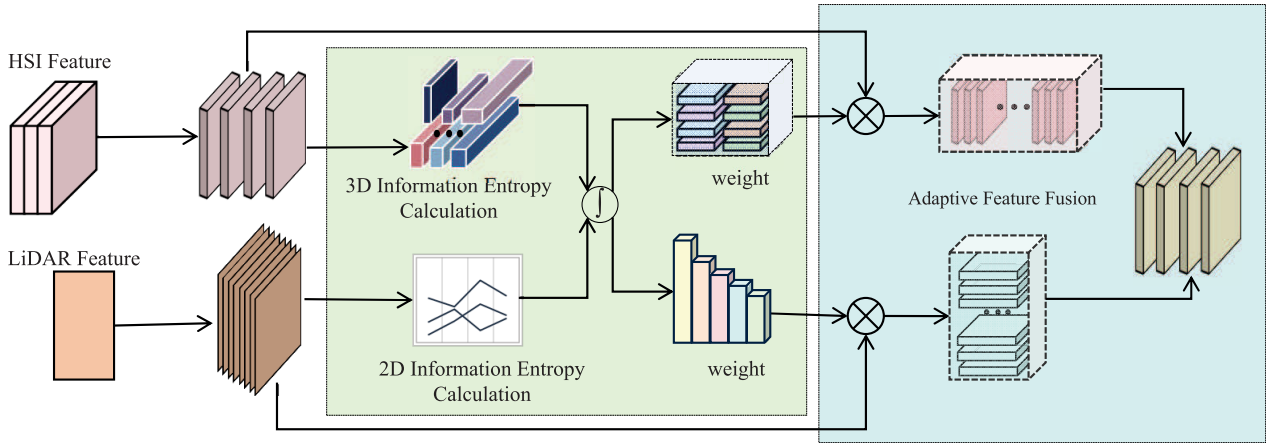


Fig. 4. Architecture of EB-DDFM.

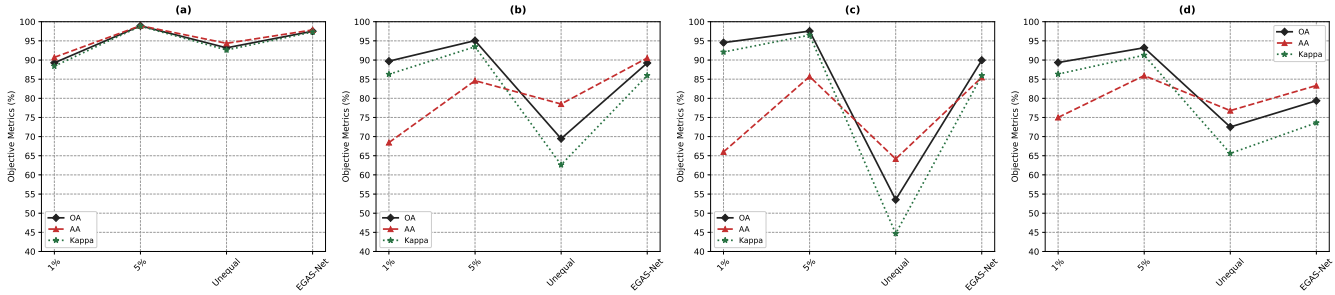


Fig. 5. Performance curves of OA, AA, and Kappa under class imbalance settings on four datasets. (a) Houston. (b) MUUFL. (c) Augsburg. (d) Yancheng.

structural fragmentation or statistical loss defects of traditional fixed-size inputs caused by scale mismatch. The model fully preserves the macrospatial layout and topological relationships of core land-cover types—including "road," "buildings shadow," "residential areas," "forest," "water," and "bare soil" across diverse scenes, such as urban, commercial, industrial, and complex wetland environments, successfully avoiding morphological distortions or large-scale displacements. The EB-DDFM leverages the entropy ratio as a quantitative indicator of feature reliability for adaptive weight adjustment and coupling, and compared to methods such as M2FNet, HCT-Net, and DFFNet, this entropy-guided mechanism allows features with high information content to play a dominant role. The proposed model successfully achieves deep optimization and integration of multisource heterogeneous features, allowing homogeneous land-cover regions in each scene to present more cohesive feature clustering and clearer, more natural global visual consistency from a macro perspective. At the local detail level, the proposed model significantly enhances the representation capability of fine-grained features through the adaptive information entropy cropping technique, overcoming the limitations of traditional fixed-size inputs. For instance, in local regions of the MUUFL dataset, compared to the frequent edge-confusion phenomenon where mPMCL and MMD-MLP models misclassify "road" as "yellow curb," our model achieves precise recognition of the "road"

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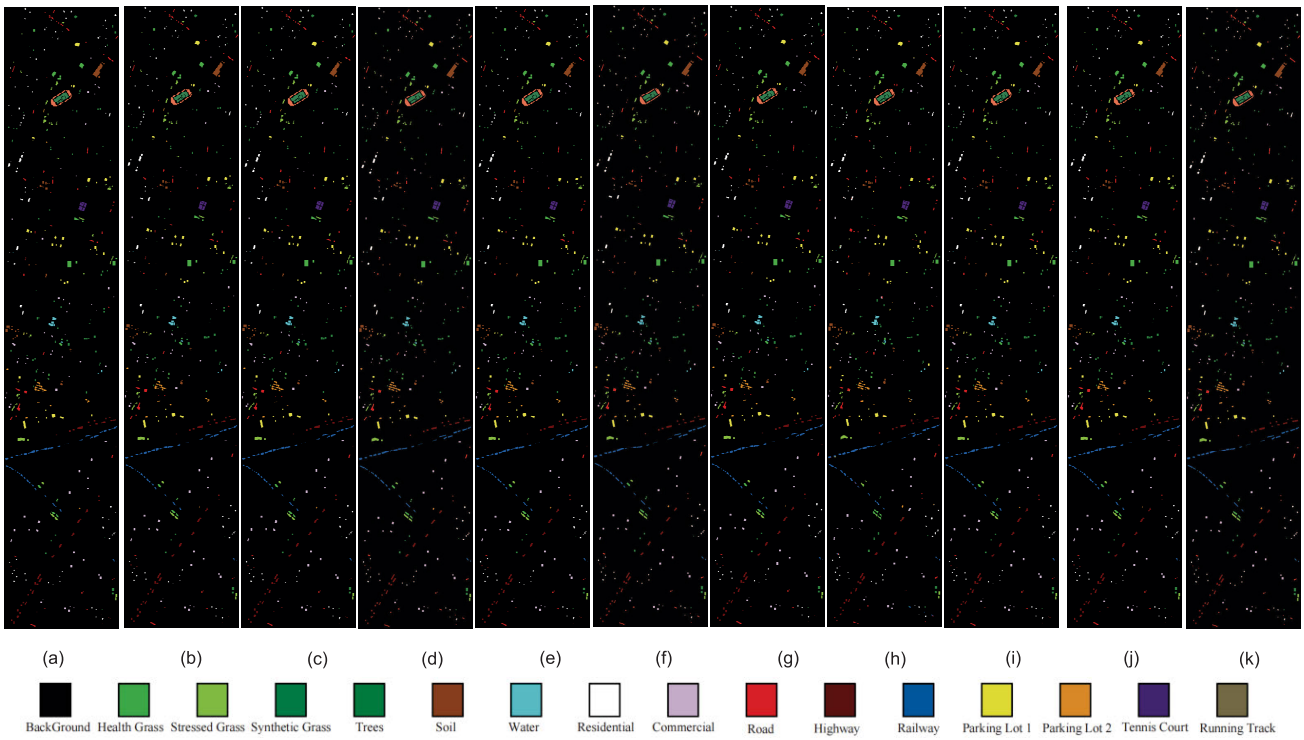


Fig. 6. Classification maps by different methods on the Houston2013 dataset. (a) Ground-truth map. (b) MHST (96.19%). (c) HCT-Net (94.02%). (d) DFFNet (85.89%). (e) FDNet (95.91%). (f) MCFNet (95.43%). (g) MSA-GCN (76.04%). (h) M2FNet (92.26%). (i) mPMCL (92.99%). (j) MMD-MLP (79.53%). (k) EGAS (97.52%).

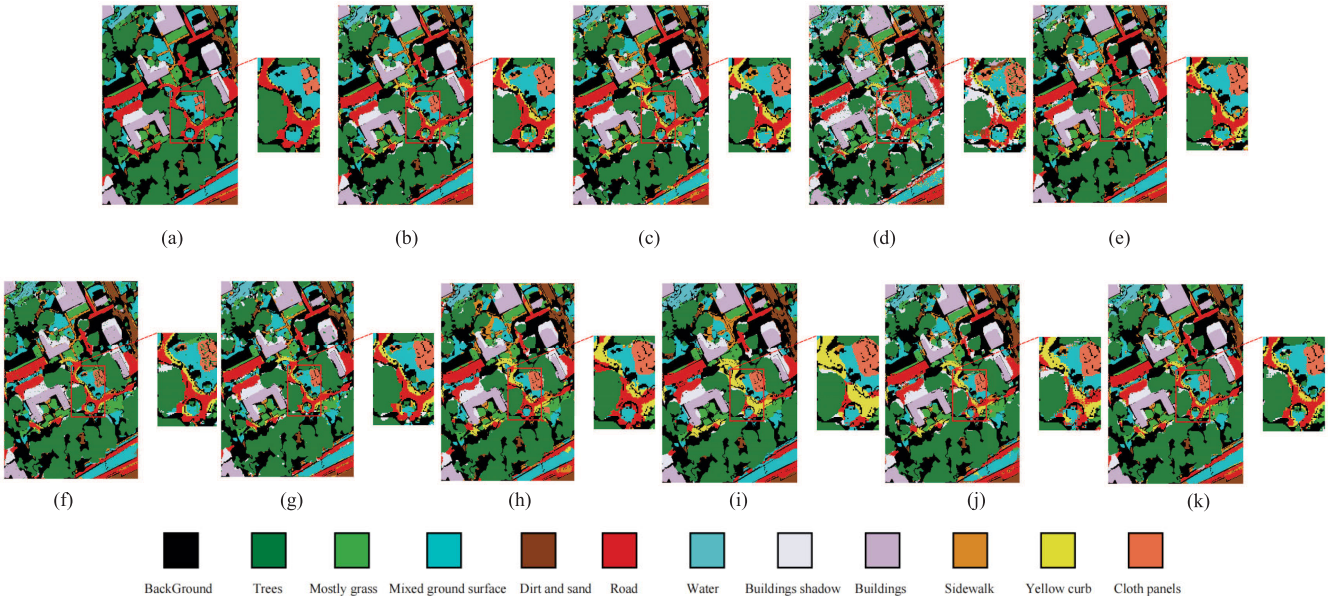


Fig. 7. Classification maps by different methods on the MUUFL dataset. (a) Ground-truth map. (b) MHST (88.71%). (c) HCT-Net (84.20%). (d) DFFNet (75.04%). (e) FDNet (88.12%). (f) MCFNet (87.46%). (g) MSA-GCN (83.30%). (h) M2FNet (81.51%). (i) mPMCL (87.26%). (j) MMD-MLP (80.44%). (k) EGAS (89.22%).

category. Furthermore, in complex scenarios like Augsburg, it is capable of more sensitively capturing local variations in fine-grained features, such as "residential area" and "commercial areas," with virtually no artifact noise. In summary, while maintaining the integrity of the overall layout, this design significantly improves local detail accuracy, effectively satisfying the requirements of macroanalysis and precise recognition in practical remote sensing applications.

3) Intermediate Visual Comparison: To achieve a more intuitive visual comparison, we use the t-SNE method to visualize the feature distributions of different models on the MUUFL dataset. Notably, this visualization method clearly

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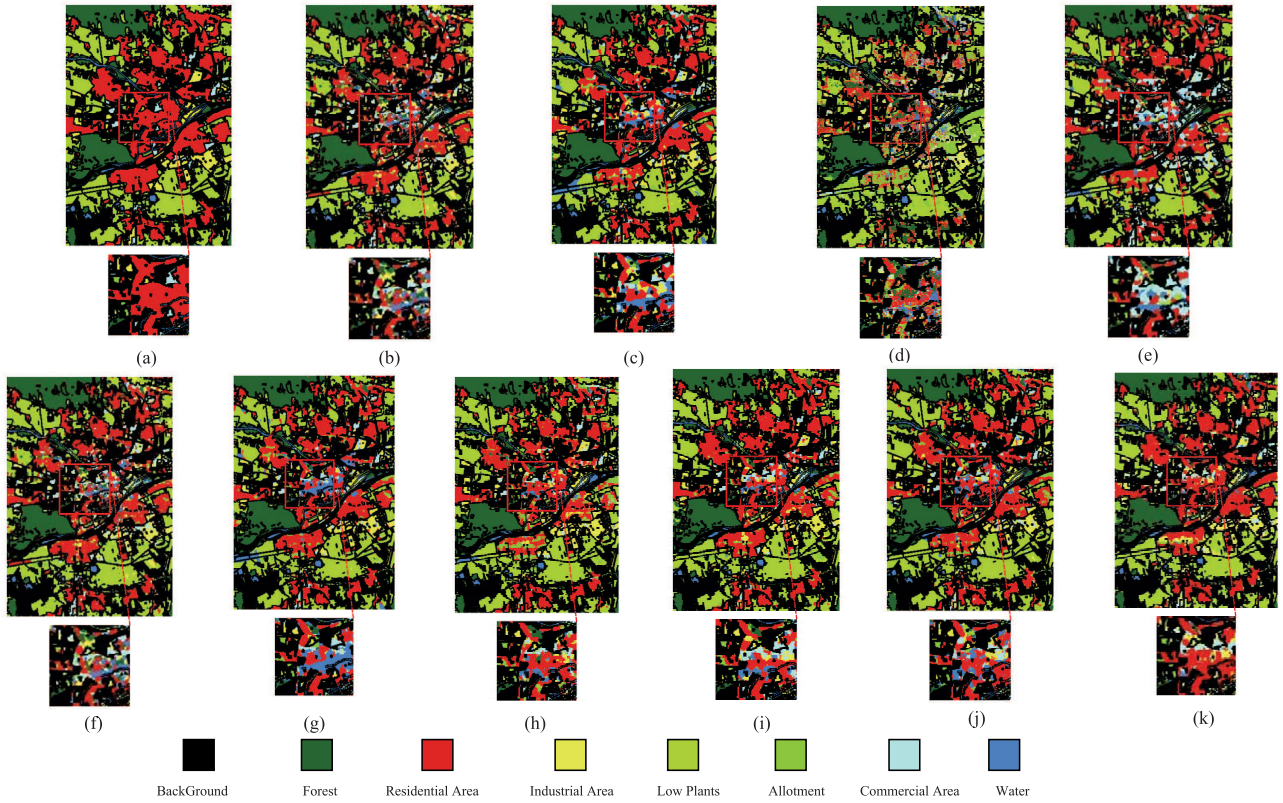


Fig. 8. Classification maps by different methods on the Augsburg dataset. (a) Ground-truth map. (b) MHST (86.16%). (c) HCT-Net (76.99%). (d) DFFNet (61.85%). (e) FDNet (82.30%). (f) MCFNet (82.83%). (g) MSA-GCN (57.63%). (h) M2FNet (85.99%). (i) mPMCL (87.80%). (j) MMD-MLP (77.32%). (k) EGAS (89.94%).

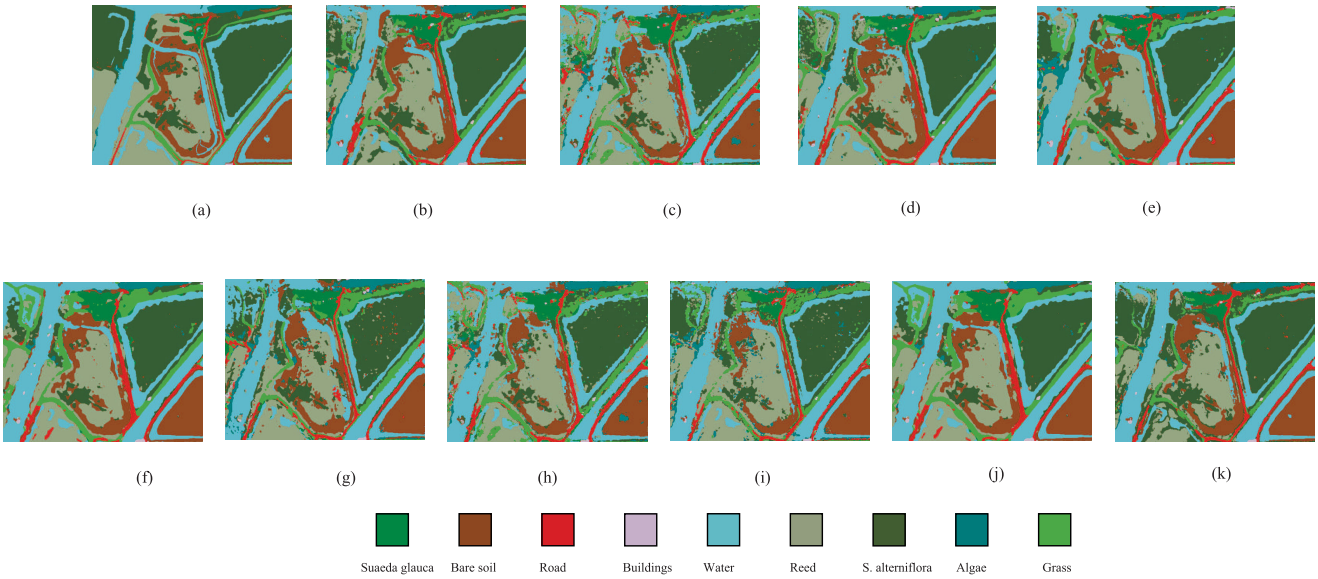


Fig. 9. Classification maps by different methods on the Yancheng dataset. (a) Ground-truth map. (b) MHST (73.33%). (c) HCT-Net (59.53%). (d) DFFNet (70.32%). (e) FDNet (76.65%). (f) MCFNet (73.46%). (g) MSA-GCN (69.27%). (h) M2FNet (72.97%). (i) mPMCL (73.57%). (j) MMD-MLP (68.09%). (k) EGAS (79.36%).

shows that the feature distribution characteristics are highly consistent with the quantitative analysis results. As shown in Fig. 10, our model demonstrates significant advantages in feature clustering and category differentiation. Data points from different categories are more clearly separated, while data points within the same category are more tightly clustered. This indicates that the model can better represent the similarity of samples within the same category and differentiate samples

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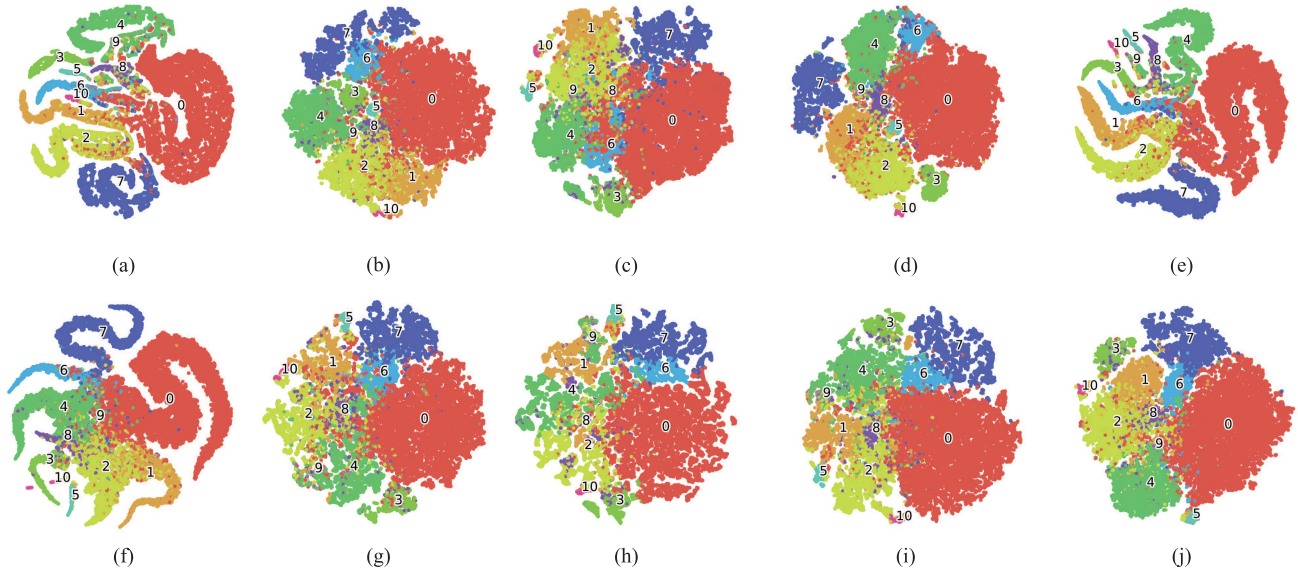


Fig. 10. t-SNE visualization of feature distribution in different methods on the MUUFL dataset. (a) MHST. (b) HCT-Net. (c) DFFNet. (d) FDNet. (e) MCFNet. (f) MSA-GCN. (g) M2FNet. (h) mPMCL. (i) MMD-MLP. (j) Ours.

TABLE V
COMPARISON OF COMPUTATIONAL COMPLEXITY AND TESTING TIME (S) OF DIFFERENT METHODS ON THE AUGSBURG DATA

Methods	MHST	HCT-Net	DFFNet	FDNet	MCFNet	MSA-GCN	M2FNet	mPMCL	MMD-MLP	EGAS
Trainable Params	336k	429k	2.28M	345.86k	190k	620k	150k	40.41M	23.61M	132k
GFLOPs	1.81	7.93×10^{-3}	3.86	11.68	1.83	0.02	0.08	0.04	1.01	0.92

of different categories more effectively. The MHST, MCFNet, and MSA-GCN models exhibit overlapping between different categories and have deficiencies in feature extraction and category differentiation capabilities. The M2FNet distribution is relatively dispersed and fails to form a good clustering structure. Moreover, there is some overlap between categories, resulting in room for improvement in representing similarities within the same category and differentiating between different categories. The mPMCL model shows severe category mixing, with samples from different categories infiltrating each other and unclear cluster boundaries. In contrast, our model achieves clearer separation of data points from different categories, with data points of the same category clustered more tightly. This demonstrates that the model can better represent the similarity of samples within the same category and differentiate samples of different categories more effectively.

4) Computational Complexity Analysis: Table V presents the computational complexity of different methods on the Augsburg dataset, including the number of trainable parameters in the backpropagation phase and giga floating-point operations per second (GFLOPs). The data shows that most comparative methods exhibit significant differences in parameter scale: for instance, mPMCL and MMD-MLP have a large number of trainable parameters due to their numerous deep branches. In contrast, lightweight networks based on convolution or Transformer, such as MHST, HCT-Net, and DFFNet, have parameters concentrated in the range of 150k–40.41 M. For the GFLOPs, FDNet alleviates part of the computational burden through DWT-based feature decomposition. However, its computational complexity still reaches 11.68 GFLOPs. M2FNet and mPMCL achieve extremely low computational costs, with GFLOPs values ranging from 0.04 to 0.08, but their classification performance is inferior to that of the proposed method. Although the proposed EGAS-Net requires 0.92 GFLOPs, it consistently achieves the best classification results across all evaluation metrics. This not only avoids the excessive compression seen in methods like FDNet and DFFNet, but also eliminates redundant computations from traditional multiscale convolutions. Regarding trainable parameters, our model has the smallest number among all methods, further reducing memory usage and inference cost significantly through parameter sharing and lightweight branch design. In summary, our model offers advantages in both parameter count and computational cost, achieving efficient lightweight design. Furthermore, our model delivers the best performance in balancing classification accuracy and model lightweightness. This further verifies that the improvement in accuracy results from EGSS's adaptive selection of informative context size, not from greater model complexity.

5) Proof of Effectiveness of the EGSS Module: The EGSS module we proposed has a plug-and-play feature. After being generalized by the model, this module demonstrates a significant improvement.

The proposed EGSS module features a plug-and-play characteristic, meaning that this module can be seamlessly integrated into existing HSI and LiDAR joint classification frameworks without requiring extensive modifications to the original model architecture. As shown in Table VI, after the model completed generalization training on the Houston2013, MUUFL, Augsburg, and Yancheng datasets, this

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TABLE VI
OA VALUES OF ALL METHODS AFTER ADDING THE EGSS MODULE

Methods	MHST	HCT-Net	DFFNet	FDNet	MCFNet	MSA-GCN	M2FNet	mPMCL	MMD-MLP
Houston2013	96.59(+**0.40**)	94.32(+**0.30**)	86.12(+**0.23**)	96.22(+**0.31**)	95.99(+**0.56**)	76.63(+**0.59**)	92.81(+**0.55**)	93.22(+**0.23**)	79.91(+**0.38**)
MUUFL	89.11(+**0.40**)	84.82(+**0.62**)	75.64(+**0.60**)	88.62(+**0.50**)	87.86(+**0.40**)	83.82(+**0.52**)	81.99(+**0.48**)	87.96(+**0.70**)	81.03(+**0.59**)
Augsburg	86.78(+**0.62**)	77.33(+**0.34**)	62.22(+**0.37**)	82.97(+**0.67**)	83.27(+**0.44**)	58.20(+**0.57**)	86.33(+**0.34**)	88.10(+**0.30**)	79.02(+**1.70**)
Yancheng	73.88(+**0.55**)	60.02(+**0.49**)	71.00(+**0.68**)	76.87(+**0.22**)	74.11(+**0.65**)	70.00(+**0.73**)	73.29(+**0.32**)	74.00(+**0.43**)	69.11(+**1.02**)

TABLE VII
EFFECTS OF DIFFERENT MODULES ON CLASSIFICATION RESULTS OF EGAS

EGSS	EB-DDFM	DDFF	Houston2013	MUUFL	Augsburg	Yancheng
✔	✗	✗	96.10	86.57	87.21	77.02
✗	✔	✗	94.12	85.03	86.03	75.12
✗	✗	✔	94.22	85.23	86.67	74.77
✗	✔	✔	95.98	86.72	88.15	76.58
✔	✗	✔	97.06	87.04	88.47	78.52
✔	✔	✗	97.16	89.18	89.10	78.11
✔	✔	✔	97.52	89.22	89.94	79.36

module exhibited significant improvements in classification accuracy metrics. Compared with mainstream methods that use fixed cropping sizes, OA consistently improved. Its core advantage lies in its ability to adaptively determine the optimal local context size for each type of land-cover target based on information entropy. This operation maximizes the extraction of discriminative features while minimizing the interference of redundant background noise. This generalization capability validates the robustness and practical value of the module in real remote sensing image classification tasks, making it particularly suitable for scenarios with significant differences in structural complexity and information richness among various types of land cover.

D. Ablation Studies

This section presents ablation experiments designed to evaluate the efficacy of various components. The experimental results are shown in Table VII.

1) Effectiveness of EGSS: To verify the superiority of the EGSS, we design two modified variants to comparatively evaluate the impact of this module on model performance. For the first variant, we remove the EGSS module and replace it with the conventional fixed-size input strategy, in which both HSI and LiDAR data are cropped to a uniform dimension before being fed into the network. For the second variant, we retain only the EGSS module while adopting a traditional fusion and classification pipeline for subsequent processing. We conduct experiments across the four datasets to compare the performance discrepancies among the complete EGAS-Net, the model without the EGSS module, and the model equipped solely with the EGSS module. Our fusion network achieved better classification results on all four datasets, reflected by higher OA and AA values. The proposed EGAS-Net can fully exploit the effective information of different classes in the data, which plays a positive role in improving classification performance. This further validates the effectiveness of the proposed EGSS and highlights its important role in extracting data information. Furthermore, the adaptive size selection mechanism, rather than any expansion of computational capacity, is responsible for the performance gains. As a nonparametric information-theoretic optimization framework, EGSS operates before feature learning and introduces no additional trainable parameters. Experimental results in Tables V and VI demonstrate that integrating EGSS consistently improves accuracy even while preserving the original model capacity, confirming that the enhancements are driven by the determination of the optimal context size. By integrating this mechanism, the model can better utilize the complex information contained in diverse feature datasets, thereby achieving more accurate and reliable classification performance.

2) Effectiveness of Information Entropy Weight in the EB-DDFM Module: To verify the superiority of the information entropy weight embedded in the EB-DDFM module, we develop two variant architectures of this module for comparative evaluation to quantify its impact on classification performance. For the first variant, we eliminate the information entropy weighting mechanism within EB-DDFM: neither the 3-D spatial entropy and frequency-domain entropy of HSI nor the 2-D spatial entropy and frequency-domain entropy of LiDAR are calculated and extracted, the corresponding entropy-driven adaptive fusion strategy is discarded without the support of information entropy, and only a simple equal-weight fusion method with a fixed weight of 0.5 is adopted for spatial and frequency information; meanwhile, we construct another ablation variant that retains merely the EB-DDFM module and removes the other two modules to implement basic weighted fusion of spatial features. Experiments were conducted on four datasets, and our model achieved higher accuracy. The proposed Information entropy weight in the EB-DDFM module can fully utilize the information entropy of spatial and frequency features of the data, thereby balancing the relationship between spatial and frequency information. This further validates the effectiveness of the proposed Information entropy weight in the EB-DDFM module and highlights its important role in extracting data information. By integrating this mechanism, the model can better employ information entropy to achieve adaptive fusion of spatial and frequency features.

3) Effectiveness of EB-DDFM: To verify the superiority of the EB-DDFM, we design two modified model variants

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to quantitatively measure how this module affects overall classification performance: one variant discards the entire frequency-domain branch and only leverages the spatial branch to complete land-cover classification, while the other variant adopts the complete dual-branch framework that jointly exploits both spatial and frequency information for multimodal classification. The comparative results further fully validate the effectiveness of the proposed EB-DDFM module. Unlike single spatial-branch structures that merely capture local pixel distribution and geometric contour information, EB-DDFM establishes independent feature extraction and adaptive entropy-guided coupling pipelines for both spatial and frequency domains for HSI and LiDAR data separately. In the spatial domain, it excavates fine-grained land-cover texture, elevation fluctuation, and spatial topological correlation, while in the frequency domain, it captures global energy distribution, periodic structural variation, and subtle interclass difference hidden in frequency components. The module organically integrates the complementary merits of spatial structural details and frequency global statistical characteristics via entropy-weighted adaptive fusion, which effectively suppresses redundant cross-modal information interference, highlights discriminative feature components with high information entropy, and substantially strengthens the feature representation capability of heterogeneous HSI–LiDAR multimodal data for subsequent precise land-cover discrimination.

V. Conclusion

In this article, an EGAS-Net for hyperspectral–LiDAR classification is proposed. Aiming at the differences in spatial feature complexity among different land-cover classes, a class-specific patch size selection strategy is designed, and a spatial–frequency adaptive fusion module is combined to improve the feature representation ability of multimodal data. In terms of data preprocessing, an entropy-driven adaptive patch size strategy is developed. By calculating the entropy distribution of HSI and LiDAR data under different patch sizes, the optimal patch size with a higher mean entropy value and more stable standard deviation is selected for each land-cover class, which effectively solves the problem that fixed patch sizes cannot adapt to the spatial feature scales of different classes. To better coordinate spatial and frequency information, we extract 3-D spatial information entropy and frequency-domain entropy for HSI, and 2-D spatial information entropy and frequency-domain entropy for LiDAR. Spatial information entropy focuses on the spatial-domain distribution characteristics of data, while frequency-domain entropy focuses on the frequency-domain characteristics. According to their respective features, an adaptive coupling method is used for fusion and interaction to achieve better integration of spatial and frequency information. Finally, joint classification is performed through a dual-branch fusion structure of HSI and LiDAR. The current research has achieved competitive performance compared with other methods on multiple public datasets, but there is still room for improvement. In the future, we can further study the information entropy processing for different frequency information. Combined with the quantization advantages of entropy features, an entropy-guided multidimensional attention fusion module and cross-branch and cross-level feature interaction structures can be designed to realize the dynamic collaboration of channel, spatial, and frequency multidimensional attention and real-time bidirectional interaction of dual-branch features. We will further explore lightweight methods for entropy calculation to reduce the computational overhead in the patch size search stage. In addition, a self-supervised learning strategy can be combined to optimize the selection of the optimal class-specific patch size using unlabeled data, so as to further improve the classification performance in small-sample scenarios.

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